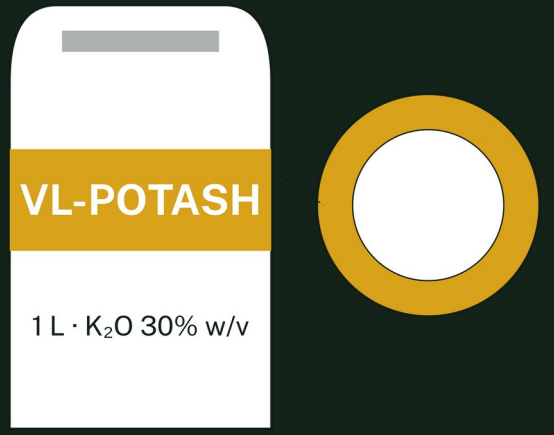
DEALER TRAINING · NOVEMBER 2026

VL-Potash Liquid



VL-Potash Liquid carries 30 g of K_2O in every 100 ml, applied on the leaf or through the drip line at the stages the crop plan gives.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

VL-Potash Liquid is potash in a bottle: K_2O 30 g in every 100 ml, 300 g in a litre, made so it stays in solution in alkaline water. The crop uses potash to fill its grain, cane, tubers and fruit. In 2024-25, potash was only 1% of all the fertiliser nutrients applied in Pakistan (NFDC), a figure also printed on VAN's Crop Force bag. A liquid can be put on at the stage the crop needs it, on the leaf or through the drip line, instead of months before. The crop plans give: wheat, 1 bottle per acre at early growth and 1 at grand growth, sprayed; sugarcane, 3 bottles at maturity; tomato, 3 bottles, 1 after each picking; guava, 1 at flowering and 1 in December; mango, 1 bottle in the irrigation water at grand growth. It adds potash in grams at the stages the crop plan gives, mostly when the crop is filling. The season's main potash is the basal fertiliser applied to the soil at land preparation.

The 4 points it must cover

1. The bottle: 1 L, K_2O 30% w/v, PS 5469-2020. 300 g of K_2O in every litre. 3 L, 20 L and 200 L packs are on the price sheet.
2. The nutrient: potash moves sugar into the grain, cane, tuber or fruit.
3. The stage: the stages your crop's plan gives, mostly when the crop is filling. It goes on the leaf or through the drip line, at the plan rate.
4. The limit: it supplies grams when demand is highest, not the season's potash. The basal potash is applied at land preparation.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the bottle or pack, in the published crop plan or in the named evidence on the slides that follow.

THE BOTTLE OR PACK

What the bottle says, and what it is registered as.

1 L

the bottle. 300 g of K_2O in every litre, from the declared analysis. 3 L, 20 L and 200 L packs are on the price sheet.

K_2O 30%

w/v: 30 g in every 100 ml. The crop plan rows give it as "Potash 30%".

pH-regulated

phosphorus and boron are added to hold the pH of the solution, as VAN's product data says. They are named for that job, not as main nutrients, and no figure is given for them.

2 routes

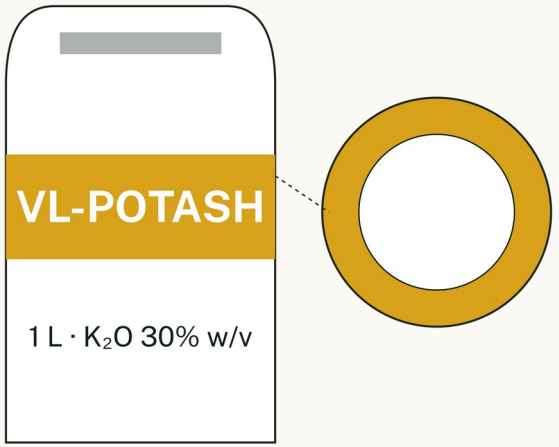
foliar and fertigation, as VAN names them. The crop plan row says which for your crop.

1%

of all the fertiliser nutrients applied in Pakistan in 2024-25 was potash (NFDC). In 2023, Pakistan applied about 83 kg of nitrogen for every 1 kg of K_2O (FAO).

Drone grade

the drone grade of VAN's potash is a separate product, with its own deck and its own rate.



Registered as	Pakistan Standard	Manufacturing licence
VL-Potash Liquid	PS 5469-2020	CM/L-2943/2023

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“I put the potash in at sowing. When the crop started to fill it looked short again. Can I give potash to a standing crop?”

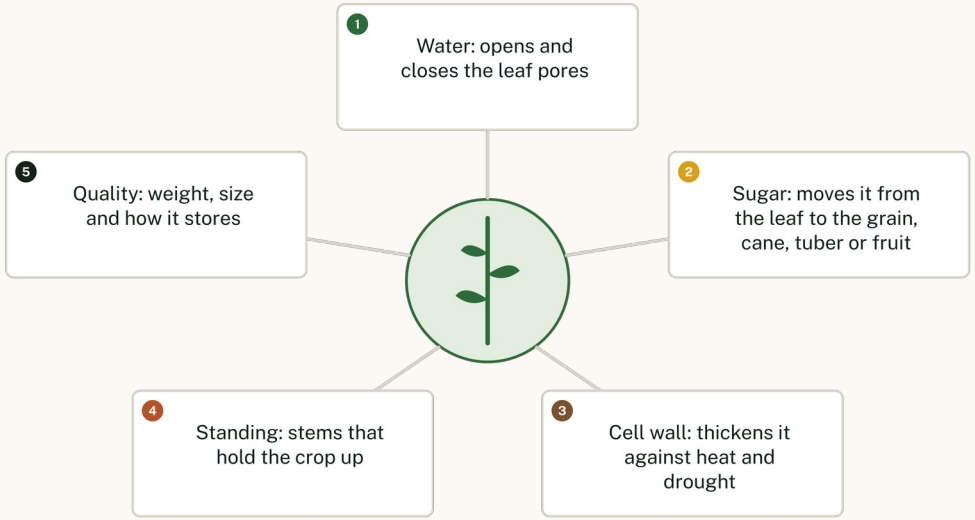
The grower applied the basal potash. He is asking whether potash can be applied at the stage the crop needs it, weeks after the bag went in. The answer has 3 parts: what potash does when the crop fills, why a liquid can be applied when demand is highest, and how it reaches the crop on the leaf or through the drip line.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

The crop uses potash to fill its grain, cane, tubers and fruit.

Potash controls the movement of water and sugar in the plant. It opens and closes the pores in the leaf, moves the sugar the leaf makes into the grain, the cane, the tuber or the fruit, and thickens the cell wall against heat and drought. A crop short of it shows scorched edges on the older leaves, weak stems, light grain and small fruit that stores badly. Because so little potash is applied, most crops in Pakistan fill their grain or fruit mainly from potassium already in the soil.



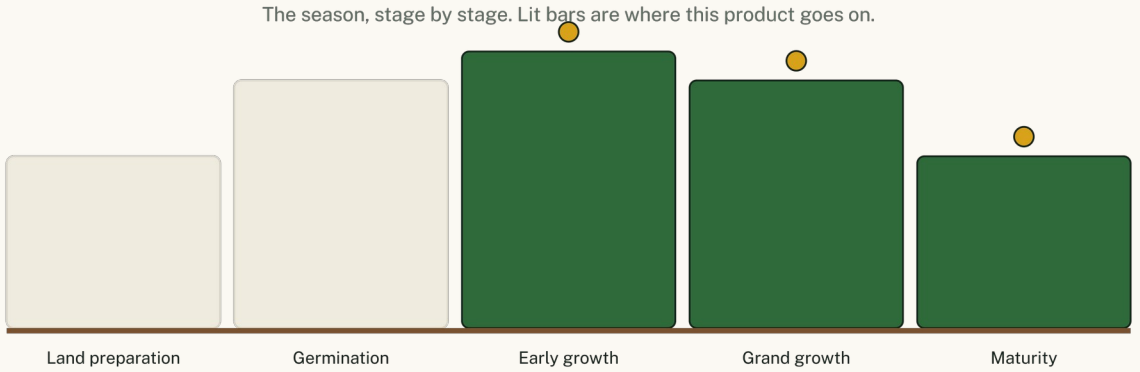
The 5 functions of potash, as VAN's 17-nutrients page gives them. The drawing shows the idea; it is not a measurement.

Say it like this: "Your basal fertiliser fed the crop's growth. This bottle adds potash at the stages the crop plan gives, mostly when the crop fills."

ANSWER 2 OF 3

A liquid can be applied at the peak of demand, not months before it.

A basal bag goes in at land preparation and the crop draws on it for the season. Potash demand peaks when the crop fills, and a liquid can be put on in those weeks and repeated when the crop needs more. The crop plans apply it at stages of high demand: wheat at early growth and grand growth, sugarcane at maturity, tomato after each picking, guava at flowering and again in December, mango at grand growth. The liquid supplies potash when demand is highest; the basal fertiliser supplies the potash the crop uses through the season.



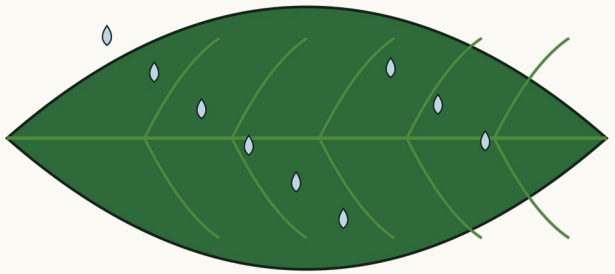
The 5 stage names on the drawing are the ones VAN's crop plans use. The crop plan slide shows where VL-Potash Liquid goes for each crop.

Say it like this: "The bag in the ground supplies the season's potash. The bottle adds potash at the stages the crop plan gives, mostly in the weeks the crop is filling."

ANSWER 3 OF 3

VL-Potash Liquid goes on the leaf or through the drip line, not into the soil as a granule.

On calcareous soil above pH 8, potash released by a granule passes through the soil before the roots take it up. A spray puts potash on the leaf, where it is taken up directly while the droplet is wet; a fertigation run carries it in the water to the wet root zone. In alkaline irrigation water, potash without pH regulation would not stay dissolved, so VAN's formulation contains phosphorus and boron to hold the pH. The product is fully soluble on both routes and leaves no residue in a drip emitter, as VAN's product data says.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

The drawing shows the leaf taking up the droplet. The method slide describes the fertigation route.

Say it like this: "It goes on the leaf or through the drip line, at the stage the crop needs it, instead of as a granule months before."

THE EVIDENCE

What is verified for VL-Potash Liquid, and what is not.

No trial of VL-Potash Liquid is on VAN's record. A dealer may quote each item below only as the right-hand column allows.

Item	What it says	How a dealer may use it
The bottle	VL-Potash Liquid, K_2O 30% w/v (300 g/L), with phosphorus and boron added for pH regulation of the solution; 1 L. 3 L, 20 L and 200 L on the price sheet.	As fact. Phosphorus and boron are named for their job, with no figure.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-2943/2023.	As fact: quote the number.
The crop plan rows	8 rows in 6 crop plans: 1 bottle per acre, spray, on wheat 2 times and guava 2 times (guava at 1 L per 100 L of water); 3 bottles, spray, on sugarcane and ratoon sugarcane at maturity; 3 bottles, spray, on tomato, 1 after each picking (3 pickings at most); 1 bottle by fertigation on mango.	As the published rate, as printed.
National potash use	In 2024-25, potash was 1% of all the fertiliser nutrients applied in Pakistan (National Fertilizer Development Centre, NFDC; also printed on the Crop Force bag). In 2023, Pakistan applied about 83 kg of nitrogen for every 1 kg of potash (K_2O) (Food and Agriculture Organization, FAO).	As fact.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	87.8% of samples below 180 ppm available potassium; mean 132 ppm. 69.1% above pH 8.	As fact about the soil. Never as product performance.
Published research, not VAN trials	Sindh banana (Memon et al. 2010): an improved practice, with far more potash than the farmers' own practice, raised yield. Multan cotton (Hussain, Ali and Gardezi 2021): higher soil potassium reduced boll shedding.	Spoken, as research on potash in the soil. Never as a VL-Potash Liquid result.
Not on file	A VL-Potash Liquid trial; a yield, sugar-recovery or quality figure; a drone rate; a Pakistani uptake curve for potato or banana.	Not said until VAN names a source.

Sources: the bottle; the Pakistan Standard and licence register; the crop plans on van.com.pk; the VAN farmer price sheet, Aug 2026; the Punjab soil survey (770,160 samples, about 2016 to 2018); Memon et al. 2010; Hussain, Ali and Gardezi 2021.

The 4 packs, in grams of K_2O and in rupees.

Prices are the delivered farmer prices from the VAN farmer price sheet, Aug 2026. K_2O is from the declared analysis, weight per volume: 30% is 300 g in a litre.

VL-Potash Liquid, K_2O 30% w/v	Volume	K_2O	Rupees	Rupees per litre
1 bottle	1 L	300 g	1,250	1,250
1 pack	3 L	900 g	3,000	1,000
1 pack	20 L	6.0 kg	18,500	925
1 drum	200 L	60 kg	175,000	875

A bottle is 300 g of K_2O on the leaf or in the drip line, for Rs 1,250. The 3 L pack holds the full sugarcane dose of 3 litres per acre. The 20 L and 200 L packs are for a dealer or a large farm; the grower buys the pack that matches the litres his plan needs.

VL-Potash Liquid 1 L Rs 1,250, 3 L Rs 3,000, 20 L Rs 18,500, 200 L Rs 175,000: the VAN farmer price sheet, Aug 2026. 300 g = 1,000 ml × 30 g per 100 ml; 6.0 kg = 20 L × 300 g. Rupees per litre = price ÷ litres.

What the crop plans apply, per acre, per season.

Bottles per acre are from the published crop plans, K_2O from the declared analysis and rupees from the VAN farmer price sheet, Aug 2026, in 1 L bottles.

Crop	Stage and method, as printed	Bottles	K_2O	Rupees per acre
Wheat	Early growth, spray; grand growth, spray	2	600 g	2,500
Sugarcane	Maturity, spray	3	900 g	3,750 (3,000 as the 3 L pack)
Sugarcane ratoon	Maturity, spray	3	900 g	3,750 (3,000 as the 3 L pack)
Tomato	Fruiting and picking phase, spray, 1 bottle after each picking, 3 pickings at most	3	900 g	3,750
Mango	Grand growth, fertigation	1	300 g	1,250
Guava	October (flowering starts) and December, foliar, 1 L per 100 L water	2	600 g	2,500

On wheat the grower spends Rs 2,500 per acre to put 600 g of K_2O on the leaf across the 2 stages the plan names. The guava crop plan is the only one that gives the water volume: 1 L per 100 L. Tomato takes 1 bottle after each picking, 3 at most in the season.

Rows from the crop plans on van.com.pk, as printed. VAN's wheat cost plan carries the same 2 L per acre at Rs 1,250 a litre. Sugarcane in 1 L bottles: 3 × Rs 1,250 = Rs 3,750 per acre. The 3 L pack, at Rs 3,000 on the same sheet, costs less.

On an acre of wheat, the harvest removes about 40 kg of K_2O; the 2 bottles add 600 g.

The table assumes a wheat yield of 40 maunds per acre (1.6 tonnes) and shows the potash this harvest removes, from the IPNI removal tables used on van.com.pk, beside the 2 sprays the crop plan gives.

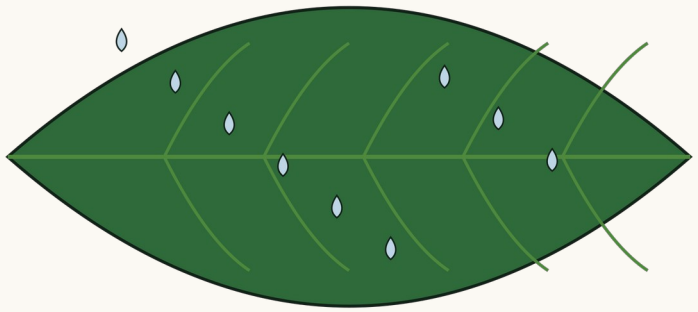
Wheat, 1 acre at 40 maunds	K_2O per tonne of grain	K_2O at 1.6 t	
Grain taken off	4.8 kg	7.7 kg	
Straw (bhusa) taken off too, per tonne of grain	20 kg	32 kg	
Grain and straw together	**24.8 kg**	**40 kg**	
VL-Potash Liquid, 2 bottles, early and grand growth, puts on		**0.6 kg**	

The 2 bottles put 600 g of K_2O on the leaf at the 2 stages when the crop builds its tillers and leaves. Using the IPNI removal tables, a 40 maund wheat crop is estimated to remove about 40 kg of K_2O per acre in the grain and the bhusa (straw) together. The bottles add potash when the crop needs it most. The season's potash comes from the basal fertiliser and from potassium already in the soil. In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), 87.8% of samples tested below 180 ppm available potassium.

Removal per tonne: the IPNI removal tables (metric 01/15; wheat grain K_2O 4.8, straw 20 per tonne of grain). 40 maunds × 40 kg = 1.6 t. 2 × 300 g = 600 g (the wheat rows). Soil: the Punjab soil survey, about 2016 to 2018.

HOW IT IS APPLIED

Spray it on the leaf (water first, then the bottle, morning or evening) or run it through the drip line.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Spray: water first, product second

Fill the tank with clean water, add the bottle, keep it agitated. The guava crop plan gives its rate: 1 L per 100 L of water. VAN's general guidance is 100 to 200 L of water per acre; the rate includes the water volume, and halving the water doubles the concentration on the leaf. Spray a fine, even fan, to cover without run-off, morning or evening, never midday.

Fertigation: dissolve, then release with the water flowing

The mango crop plan applies 1 bottle in the irrigation water at grand growth. Put it into the drum or the drip tank, stir it in and set the tap so that the drum empties slowly over the whole irrigation, not all at once. It is fully soluble and leaves no residue in the emitter, as VAN's product data says.

At the stage your crop plan gives

Wheat at early growth and grand growth. Sugarcane and ratoon, 3 bottles at maturity. Tomato, 1 bottle after each picking, 3 at most. Guava at flowering and in December. Never apply it to the soil surface, never use it in place of the basal bag, and never use these rates in a drone.

OBJECTION 1 OF 8

“1,250 for 1 litre of potash? A bag of SOP is 8,000 for 25 kg.”

What the dealer says

They do 2 different jobs. The bag goes in the ground at land preparation and the crop draws on it all season; the bottle goes on the leaf or through the drip line at the stages the crop plan gives, mostly when the crop is filling. On wheat it is 2 bottles per acre, 2,500 rupees: 1 at early growth, when tillers begin to form, and 1 at grand growth, when tillering and leaf growth reach their maximum. Together they put 600 g of K_2O on the leaf. Write down both as 2 separate lines; the crop plan uses both.

The facts behind the answer

- VL-Potash Liquid 1 L Rs 1,250; 3 L Rs 3,000; 20 L Rs 18,500; 200 L Rs 175,000 (VAN farmer price sheet, Aug 2026). SOP 25 kg is Rs 8,000 on the same sheet.
- 300 g K_2O per litre, from the declared analysis (the bottle).
- Wheat: 2 bottles per acre, early growth and grand growth, spray (the crop plans on van.com.pk).
- The basal potash is a different bag in the same crop plan.

Never say: “It gives you X more maunds”, a rupees-per-acre return, or a cost per kg of K_2O against the bag. No yield figure exists for this product.

OBJECTION 2 OF 8

“Can I skip the basal potash and spray this instead?”

What the dealer says

No. A litre is 300 g of K_2O. The crop removes far more: a 40 maund wheat crop removes an estimated 40 kg of K_2O per acre in the grain and bhusa together. The bottle adds grams of potash at the time of highest demand. The basal fertiliser supplies the season’s potash in kilograms. Keep the bag in the ground at land preparation and add the bottles at the stages the crop plan gives.

The facts behind the answer

- Wheat at 40 maunds removes an estimated 7.7 kg K_2O in the grain and about 40 kg in grain and straw together (the IPNI removal tables).
- 2 wheat bottles = 600 g K_2O.
- On wheat, sugarcane, ratoon sugarcane, tomato and mango, the crop plan also has a basal potash row.

Never say: “2 sprays and you can skip the bag.”

OBJECTION 3 OF 8

“Will it burn the leaf?”

What the dealer says

Not at the crop plan rate and the water VAN gives. The guava crop plan says 1 L per 100 L of water; for other crops, VAN’s general guidance is 100 to 200 L per acre. Burn happens when the water is cut to cover more acres with each tank, and when the spray goes on at midday. Water first, product second, keep it agitated, a fine fan to cover without run-off, morning or evening.

The facts behind the answer

- Guava row: "Foliar, 1 L per 100 L water" (the crop plans on van.com.pk).
- 100 to 200 L of water per acre, morning or evening, never midday (VAN guidance).
- Cutting the water in half doubles the concentration on the leaf.

Never say: “Use less water”, or “it cannot burn”.

OBJECTION 4 OF 8

“Can I put it in my drone?”

What the dealer says

No, not this bottle. VAN makes a low-volume drone grade of its potash, made for a 20 L tank, with its own name, its own specification and its own deck. This bottle, mixed at drone concentration, is not the drone product and can scorch the leaf. Ask VAN for the drone product and its rate; I will not work one out from this bottle.

The facts behind the answer

- VAN’s drone grade of potash is a separate product with its own specification and deck.
- For its drone products only, VAN’s standard rate is 1 L of product in a 20 L tank for each acre.
- This bottle, mixed at drone concentration, is not the drone product and can scorch the leaf.

Never say: “Use the 1 L bottle in the drone at 1 L per acre,” or any drone rate worked out from this bottle.

OBJECTION 5 OF 8

“Which stage, and how many bottles?”

What the dealer says

These are the crop plan rates. Wheat: 1 bottle at early growth and 1 at grand growth, sprayed. Sugarcane and ratoon: 3 bottles at maturity, sprayed. Tomato: 1 bottle after each picking, 3 bottles at most. Guava: 1 bottle at flowering in October and 1 in December, at 1 L per 100 L of water. Mango: 1 bottle in the irrigation water at grand growth. Do not apply it at sowing: that is the stage for the basal bag.

The facts behind the answer

- Wheat 2 × 1 bottle, spray; sugarcane and ratoon 3 bottles at maturity, spray (the crop plans on van.com.pk).
- Tomato 1 bottle after each picking, 3 at most; guava 1 bottle in October and December; mango 1 bottle by fertigation at grand growth (the crop plans on van.com.pk).
- VAN’s product page gives the stage as “Maturity”; the wheat rows add early and grand growth.

Never say: “More bottles, more yield”, or a rate for a crop with no row.

OBJECTION 6 OF 8

“Is it registered? What is the licence?”

What the dealer says

Yes. VL-Potash Liquid is registered under Pakistan Standard PS 5469-2020 for liquid nutrient supplements, and VAN makes it under manufacturing licence CM/L-2943/2023. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. On van.com.pk, enter the batch number printed on the bottle, and you can read the certificate for that batch.

The facts behind the answer

- VL-Potash Liquid: PS 5469-2020, CM/L-2943/2023 (the licence register on van.com.pk).
- Every batch is tested at VAN Lab, PNAC LAB 336, before it ships.
- A grower can read the certificate for any batch on van.com.pk by its batch number.

Never say: “This licence is valid until …” Quote the licence number only; never state an expiry date for the licence.

OBJECTION 7 OF 8

“Can it go through the drip on my other crops, not just mango?”

What the dealer says

Not at the spray rate given for your crop. VAN names 2 routes for this product, foliar and fertigation, but only the mango crop plan gives a fertigation rate. It is fully soluble and leaves no residue in the emitter. If you want to run it through the line on another crop, ask VAN for the fertigation rate for that crop rather than putting the spray bottle count into the tank.

The facts behind the answer

- VAN names foliar and fertigation as the routes for VL-Potash Liquid.
- Fertigation is given on the mango row only (the crop plans on van.com.pk).
- VL-Potash Liquid is fully soluble and leaves no residue in an emitter (VAN product data).

Never say: “Put the spray dose in the drip”, or any fertigation rate that no crop plan gives.

OBJECTION 8 OF 8

“My cane is standing tall but the mill says the recovery is low. Will this fix it?”

What the dealer says

Potash moves sugar from the leaf into the stalk, and the sugarcane crop plan applies 3 bottles per acre at maturity, the stage when this happens. Put it on as the plan says. I will not promise you a recovery figure, because VAN has not measured one for this product, and a mill result has many causes besides potash.

The facts behind the answer

- Potash moves sugar from the leaf to the grain, cane, tuber or fruit (VAN’s 17-nutrients page).
- Sugarcane and ratoon: 3 bottles at maturity, spray (the crop plans on van.com.pk).
- No sugarcane trial of VL-Potash Liquid is on VAN’s record.

Never say: “It raises recovery by X”, “it ripens the cane”, or any sucrose figure.

ITS PLACE IN THE CROP PLANS

VL-Potash Liquid in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	1 pack	Spray	1 L
Wheat	Grand Growth	1 pack	Spray	1 L
Sugarcane, February planting	Maturity	3 packs	Spray	1 L
Ratoon Sugarcane	Maturity	3 packs	Spray	1 L
Tomato	Fruiting and Picking Phase	3 packs (1 after each picking, 3 pickings at most)	Spray	1 L
Mango	Grand Growth	1 pack	Fertigation	1 L
Guava	October, flowering starts	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	December	1 pack	Foliar, 1 L per 100 L water	1 L

The rows are printed as published in the crop plans on van.com.pk. Coverage: these are all 8 rows in the 6 crop plans that name VL-Potash Liquid. Of these 8 rows, 5 say spray, 2 say foliar and 1 says fertigation. Every row uses the 1 L pack. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield, sugar-recovery, firmness or shelf-life figure. No VL-Potash Liquid trial exists.
2. That it replaces the basal potash or is the season's potash. It supplies grams when demand is highest; the basal fertiliser supplies the season's potash.
3. A drone rate for this bottle, or the name of the drone grade. The drone grade is a separate product with its own deck.
4. A disease, cold or stress-tolerance claim, or a claim about nitrogen use efficiency. Potash does thicken the cell wall, but nothing has been measured for this product.
5. A phosphorus or boron figure. They are in the bottle to hold the pH, named for that job, with no figure.
6. A water volume or a fertigation rate that no crop plan gives, as if it were a crop plan figure.

HOW A GROWER VERIFIES A BOTTLE OR PACK

Every batch is tested, and the batch number on the bottle or pack identifies its certificate.

1 · The batch number

Every bottle or pack carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any bottle or pack by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for VL-Potash Liquid shows

The certificate reports the K_2O measured in the batch, against the declared analysis K_2O 30% w/v. The batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-2943/2023 are on the bottle. Phosphorus and boron have no declared figure, so the certificate does not report them. The grower reads the certificate for his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Promise word GUARD: "the crop's shield". *English meaning · Urdu reviewed 27 Sep 2026*	فصل کی ڈھال
At the crop plan stages, mostly as the crop fills, give potash on the leaf. *English meaning · Urdu line pending review*	فصل کے پلان کے مرحلوں پر، زیادہ تر بھرائی کے وقت، پتے پر پوٹاش دیں
This is the potash for the filling weeks. The season's potash is the bag in the ground. *English meaning · Urdu reviewed 27 Sep 2026*	یہ بھرائی کے دنوں کی پوٹاش ہے، موسم بھر کی پوٹاش زمین والی بوری ہے
Water first, then the bottle. Morning or evening. *English meaning · Urdu reviewed 27 Sep 2026*	پہلے ٹینک میں پانی، پھر بوتل۔ صبح یا شام

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

VL-Potash Liquid: on the leaf or through the drip line, at the crop plan stages. Basal potash goes in at land preparation.

